
15-859(B) Project Report: Connections of Evolvability to Correlational SQ Learning

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1 Introduction

Darwinism, the theory of biological evolution was introduced by Charles Darwin in 1859. One of the core ideas from the theory is that of survival of the fittest organism determined by the environment in which the organism exists. The organism or individual aims to learn and maximize the benefit. On the other hand, the environment imposes restrictions to the learning of the organism. In molecular biology, organisms are known to take inputs from the environment via a type of protein known as *transcription factor*. This protein regulates the amount of other proteins to be produced by genes based on the inputs from the environment. The question that transcription factors impose is how the protein production is regulated? In other words, how does the regulation of proteins evolve over inputs from the environment? Obviously they cannot try all possible combinations.

In this report, we shall try to show under certain restrictions that *evolvability* is equivalent to the ubiquitous *Probably Approximately Correct* (PAC) model [5]. This is pertinent because it provides *evolvability* a quantitative theory such that one can formally provide bounds on learning via *evolvability*. For the rest of the report, we shall use the terms *representation* and *functions* interchangeably.

The report aims to convey three ideas:

- Definition of Valiant's model of *evolvability*
- The equivalence of *evolvability* to correlational statistical query
- How under Valiant's model can an evolution algorithm reach close to desired hypothesis?

2 Valiant's model of Evolvability

Let's consider some ideal function f (like a linear threshold function), and another function r which tries to approximate f . In general, we would like to assert that a class C of ideal functions are evolvable if any f in the class C satisfies two conditions:

1. Starting from function r_0 we shall obtain a sequence of functions $r_0 \Rightarrow r_1 \Rightarrow r_2 \Rightarrow \dots$ such that each r_i will be a consequence of a single step of mutation and selection from r_{i-1}
2. After some moderate number of steps i , the resultant r_i will have performance significantly higher than the performance of r_0 (r_0 can be initialized to any function).

Let's first elaborate how an algorithm will produce a subsequent representation r_{i+1} based on the current representation r_i . We shall address condition 2 in subsequent sections of the report.

2.1 Evolution Algorithm

Definition 1. As stated in [3], an evolution algorithm A is defined by a pair (R, M) where:

- R is a class of representation of functions (like linear threshold functions) over domain $X \in [-1, 1]$

- M is a randomized polynomial time algorithm that given a representation $r \in R$ as input, outputs a mutated representation $r' \in R$. M can output a finite set of representations using r , and we shall denote all such possible mutated representation of r as the neighbourhood $N(r)$. It should be noted here that M is an efficient algorithm i.e. has some polynomial running time in $n, \frac{1}{\epsilon}$, where n is the dimensional space of our linear threshold functions, and ϵ is the error rate.

2.2 Selection

There are various models describing how to make selection of a representation in work by Feldman [2], but here we shall look at a generalized version. For the selection rule, p candidate representations are created using the mutation algorithm M . For each candidate representation $r_i \in N(r)$, we associate an empirical performance defined as:

$$v(r_i) = \widetilde{\text{Perf}}_D(f, r_i) = \frac{1}{S} \sum_{j \leq S} r_i(x_j) f(x_j), \quad (1)$$

where D is the distribution over the domain X ; f is the ideal function; S is the sample size. Let t be some tolerance, then using t we shall form two disjoint sets of neighbours: 1) Beneficial set $Bene(N(r)) = \{r_i | v(r_i) \geq v(r) + t\}$, and 2) Neutral set $Neut(N(r)) = \{r_i | |v(r_i) - v(r)| < t\}$. Then, the selection algorithm will perform one of the following steps:

1. if $Bene(N(r)) \neq \emptyset$, then output $r_i \in Bene$ with probability $Pr_{N(r)} r_i / \sum_{r_i \in Bene(N(r))} Pr_{N(r)} r_i$
2. else if $Bene(N(r)) = \emptyset$ and $Neut(N(r)) \neq \emptyset$, then output $r_i \in Neut$ with probability $Pr_{N(r)} r_i / \sum_{r_i \in Neut(N(r))} Pr_{N(r)} r_i$
3. else if $Neut(N(r)) \cup Bene(N(r)) = \emptyset$, then output $\emptyset$ and stop.

Definition 2. Based on the selection mechanism, we say that evolution algorithm A is said to evolve C over D monotonically, if with probability at least $1 - \epsilon$ for every $i \leq g(n, \frac{1}{\epsilon})$, the empirical performance $\widetilde{\text{Perf}}_D(f, r_i) \geq \widetilde{\text{Perf}}_D(f, r_{i-1})$. This is true for any i since the evolution algorithm can be started at any initial representation r_0 . Note g is some polynomial function of $n, \frac{1}{\epsilon}$.

3 Statistical Query and Correlational Statistical Query

Here we will briefly introduce the statistical query model of Kearns [4]. In such a model, the algorithm has access to a statistical query oracle $Stat(f, D)$ for a target concept f and a distribution D . A query to this oracle is a function $\psi : X \times [-1, 1] \Rightarrow \{-1, 1\}$. The oracle responds with a value v such $E_D[\psi(x, f(x))] - v \leq \tau$, where τ is the tolerance to a query. An algorithm A is said to learn a class of functions C from statistical queries of some tolerance if A PAC learns C using $Stat(f, D)$. A correlational statistical query is a special case of statistical query for a correlation function. Specifically, $\psi(x, f(x)) \equiv r(x) \cdot f(x)$. Informally, SQ has a function $\psi(\cdot)$ which is dependent on the example, and CSQ looks at the correlation of the current realization $r(x)$ and the ideal function $f(x)$.

We observe that the CSQ oracle works similarly to the evolvability oracle defined in Eq. 1.

4 Evolvability of Sparse Linear Functions

Now that we have provided the definition of evolvability; informally shown the equivalence to correlational statistical query, we shall now, under certain limitations, show that linear functions are *evolvable* i.e. after certain moderate number of steps the mutation algorithm M will generate representation r_i with significantly higher empirical performance as compared to r_0 . Synonymously, r_i will be closer to the ideal function. This is based on the work by Kanade [1]

Sparse linear functions are represented as a mapping from $\mathbb{R}^n \Rightarrow \mathbb{R}$ by a vector $w \in \mathbb{R}^n$:

Definition 3. $Lin_{l,u}^k = \{x \mapsto w \cdot x | \text{sparsity}(w) \leq K, \forall i, w_i = 0 \text{ or } l \leq |w_i| \leq u\}$

Definition 4. Let $\tilde{D}$ be some distribution over $\mathbb{R}^n$, and draw $\tilde{x} \sim \tilde{D}$. A distribution D is smooth over x if it is obtained by the convolution i.e. $D = \tilde{D} * U_\Delta^n$ i.e. $\forall i, x_i = \tilde{x}_i + \eta_i$, where $\eta_i \in [-\Delta, \Delta]^n$ is chosen uniformly at random. (see Figure 1)

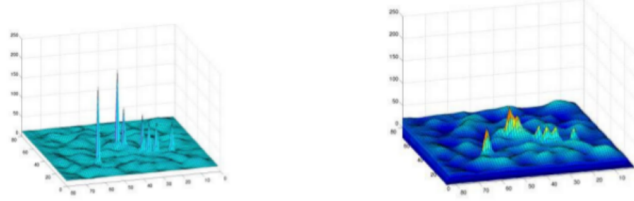
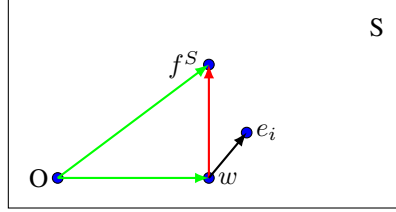


Figure 1: Smoothened distribution on the right (Image credits: Varun Kanade)

Figure 2: Adjustment Mutations



Based on the above definitions, the representation $r(x)$ for our evolutionary algorithm shall be of the form:

$$r(x_1, \dots, x_n) = \sum_i w_i x_i \quad (2)$$

4.1 Mutation Types

The mutation algorithm shall randomly perform any one of the mutation types defined below. Let non-zero variables be represented by $NZ(w) = \{i | w_i \neq 0\}$, then:

1. With probability $1/3$, do *Scaling*: Choose $\gamma \in [-1, 1]$ uniformly at random and let $w' = \gamma w$.
2. With probability $1/3$, do *Adjusting*: Set $w \leftarrow w - w_i e^i + \beta_i e^i$, for a random $i \in NZ(w)$, $\beta_i \in [-B, B]$
3. With remaining probability $1/3$ do:
 - (a) *Swapping*: Choose a pair (i_1, i_2) , where $i_1 \in NZ(w)$ is chosen uniformly at random; $i_2 \in [n] \setminus NZ(w)$ is also chosen uniformly at random. If w' is the mutated representation of w such that $w'_j = w_j$ for $j \neq i_1, i_2$. Set $w'_{i_1} = 0$ and $w'_{i_2} = \eta$, where $\eta \in [-B, B]$.
 - (b) *Adding*: if $|NZ(w)| < K$, choose $i \in [n] \setminus NZ(w)$ uniformly at random. Let w' be the mutated representation of w such that $w'_j = w_j$ for $j \neq i$, and choose $w'_i \in [-B, B]$

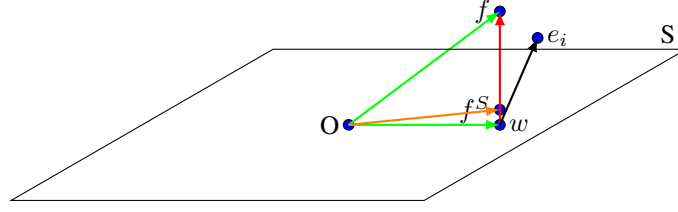
The above mutations types are applicable only if the distribution is smooth. The performance/fitness here is a quadratic function:

$$L_{f,D} = \|f - w\|^2 = \|f - f^S\|^2 + \|f^S - w\|^2, \quad (3)$$

where f^S denotes the best approximation f , and $S = NZ(w)$. Please note that the above mutation algorithm will be used multiple times, say p , to produce p neighbours of representation r . The goal of this setting of the problem is: if $\|f^S - w\|^2$ is large, then the mutation of types - *scaling* and *adjusting* shall bring w closer to f^S by reducing the expected loss. Conversely, if $\|f^S - w\|^2$ is small and $S \neq NZ(f)$, then there must be a variable in $NZ(f) \setminus S$ which when added to w by *swapping* reduces the expected loss. Thus, a beneficial mutation is produced with high probability.

Consider figure 2 such that $|S| \leq K$. Then, either w is the best representation using S , or adjustment of some w_i is an improvement. Difference is shown by the *red* line, and reduced if w moved in the direction of e_i . Similarly, in figure 3 $\|f^S - w\|^2$ is low, and if we apply *swapping* - discard low weight variable from w and add a new variable not in S (given by e_i in figure), then the expected loss will be reduced. Please note f^S is contained within the parallelogram and is the best approximation with current set of non-zero variables S .

Figure 3: Swapping Mutations



Here, we have provided informally as to why a beneficial mutation is produced with high probability. We shall now show the existence of a beneficial mutation for distribution-independent evolvability [3].

5 Proof for existence of beneficial neighborhood

¹ We shall assume that there exists some margin γ on distribution D from the target halfspace f if the current hypothesis ϕ is not ϵ -close to the optimal.

Theorem 5.1. *For every halfspace f with margin γ on an arbitrary distribution D and every $\epsilon > 0$, there exists $\phi' \in N_\alpha(\phi)$ such that:*

$$\|f - \phi'\|_D^2 \leq \max(\|f - \phi\|_D^2 - \alpha^2, \epsilon), \quad (4)$$

where $\alpha = \frac{\epsilon\gamma}{3\sqrt{n}}$

Proof. Let $f = \text{sign}(\sum_{i \in [n]} w_i x_i - \theta)$. The claim holds if $\|f - \phi\|_D^2 \leq \epsilon$. Hence, we will assume $\|f - \phi\|_D^2 > \epsilon$. The difference $|f - \phi|$ shall be 0 if both functions agree, and 2 otherwise. If $a \in [-2, 2]$, $|a| \geq a^2/2$. Hence, $E_D[|f - \phi|] \geq \epsilon/2$.

For every x via distribution D , $f(x) - \phi(x)$ will have the same sign as $f(x)$, and therefore the same sign as $\sum_{i \in [n]} w_i x_i - \theta$. Therefore,

$$E \left[(f - \phi) \left(\sum_{i \in [n]} w_i x_i - \theta \right) \right] \geq \gamma E_D[|f - \phi|] \geq \epsilon\gamma/2 \quad (5)$$

Also, by using Cauchy-Schwartz inequality we can get an upper bound:

$$E \left[(f - \phi) \left(\sum_{i \in [n]} w_i x_i - \theta \right) \right] \leq \sqrt{2} \sqrt{\sum_{i \in [0 \dots n]} E_D[(f - \phi)x_i]^2} \quad (6)$$

By squaring and combining Equation 5 and 6, we get:

$$\sum_{i \in [0 \dots n]} E_D[(f - \phi)x_i]^2 \geq (\epsilon\gamma)^2/8 \quad (7)$$

Therefore there will exist some $j \in [0 \dots n]$ which will satisfy:

$$|E_D[(f - \phi)x_j]| \geq \epsilon\gamma/\sqrt{8(n+1)} \geq \epsilon\gamma/(3\sqrt{n}) = \alpha \quad (8)$$

□

Finally, we shall show that step in the direction of x_j from ϕ will decrease the distance to f .

Theorem 5.2. *Let $\alpha' = \alpha \cdot \text{sign}(E_D[(f - \phi)x_j])$, where $\alpha = \epsilon\gamma/3\sqrt{n}$ (as above). Then:*

$$\|f - (\phi + \alpha' \cdot x_j)\|_D^2 \leq \|f - \phi\|_D^2 - \alpha^2$$

¹The proof is taken from [3], but shortened

Proof. Define

$$P_1(a) = \begin{cases} a, & \text{if } |a| \leq 1 \\ \text{sign}(a) & \text{otherwise} \end{cases}$$

Let $\phi' = P_1(\phi + \alpha' \cdot x_j)$. If for a point x , $\phi'(x) = \phi(x) + \alpha' \cdot x_j$ then:

$$f(x) - \phi'(x) = f(x) - (\phi(x) + \alpha' \cdot x_j) \quad (9)$$

when $|\phi(x) + \alpha' \cdot x_j| \leq 1$. Else, if $|\phi(x) + \alpha' \cdot x_j| > 1$ then $\phi'(x) = \text{sign}(\phi(x) + \alpha' \cdot x_j)$ and for any value $f(x) \in \{-1, 1\}$:

$$|f(x) - \phi'(x)| \leq |f(x) - (\phi(x) + \alpha' \cdot x_j)| \quad (10)$$

This implies that:

$$\|(f - \phi')\|_D^2 \leq \|f - (\phi + \alpha' \cdot x_j)\|_D^2 \leq \|f - \phi\|_D^2 - \alpha^2 \quad (11)$$

□

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